Micro I, class 18

After reviewing the proof of Pratt's theorem on risk aversion and asset demand, I suggested a definition of an *increase in risk*. For F, G in $\mathcal{L}$, G is riskier than F if

$$\int f(z)dG(z) \leq \int f(z)dF(z) \quad \text{for every concave } f \text{ on } \mathbb{R}_+.$$

The definition implies that the variance of G is greater than or equal to the variance of F. The converse is false (even if we restict attention to cdfs with the same mean).

The final issue for choice theory under uncertainty is a *subjective expected utility* (SEU) representation. In the vN-M version of expected utility, probabilities are part of the environment: it is what preferences are defined over. There is no sense in which two people can have different beliefs; and no sense of learning new things and updating beliefs.

In this version of expected utility, preferences are not define over probability distributions (over outcomes), but over *acts*, functions from unknown states of the world, a set S, into consequences. Since I take the set of states to be finite, I can describe an act by a vector, where the s-the component is the outcome that the act gives in state s. A conclusion of the theorem is that, under some axioms, there is a vN-M utility u *and* a probability distribution π on S such that act y is at least as good as act x if and only if

$$\sum_{s \in S} u(y_s)\pi(s) \geq \sum_{s \in S} u(x_s)\pi(s).$$

I have a note sketching the Anscomb-Aumann (1961) proof of their theorem characterizing this representation, but alas do not have time to discuss it.

I presented the *Keynes-Ellsberg paradox*, a thought experiment that casts doubt on the idea that choices can be rationalized by probabilities. Imagine two urns. Urn I contains 50 red and 50 white balls; Urn II contains 100 red and white, but you don't know the number of each. Presumably most would be indifferent between these two acts: (a) winning \$100 if a red ball is drawn from Urn II, nothing otherwise; and (b) winning \$100 if a white ball is drawn from Urn II, nothing otherwise. Within SEU this implies that the subjective likelihood of white in Urn II is 1/2. Yet many people strictly prefer the act (a) to act (c), winning \$100 if a red ball is drawn from *Urn I*, nothing otherwise. But in SEU, acts (a) and (c) must be indifferent, since they generate the same subjective distribution over outcomes.

On Thursday I will begin with a digression on the textbook monopoly model, summarized in my notes on the topic. You ought to be able to finish the assignment without me presenting the model in class, but I will defer the official due date by 24 hours.